FLOOR MANUFACTURING ERP
END-TO-END PROCESS & FUNCTIONAL SUPPORTING DOCUMENT

Integrated Procurement • Inventory • Quality/Laboratory • Production • Sales • Gate Management

1. Purpose and Scope

This document translates the supplied Scope of Work and the uploaded raw-material procurement, supplier, purchasing, inspection, store receiving, stock adjustment, and supplier-payment requirements into a single integrated ERP process model for Floor Manufacturing.

The design follows the source process where a Storekeeper raises a Purchase Requisition (PR), Management approves it, Purchasing creates and sends the Purchase Order (PO), the supplier delivers, Inspection/Quality Control validates the material, the Storekeeper receives accepted goods and creates a Goods Receipt Note (GRN), and Finance raises and approves a payment request using the PO, GRN and supplier invoice.

2. ERP Functional Landscape

Module	Primary responsibility	Key outputs / handoffs
Procurement Management	Supplier registration, PR, PO, approvals, procurement tracking	Approved PO, supplier commitment, procurement status
Inventory Management	Items, warehouses, stores, silos, stock balances, movements, adjustments	Available/blocked stock, stock ledger, inventory balances
Raw Material Testing & Laboratory	Inspection, test parameters/results, approval/rejection, quality records	Accepted/rejected material status; quality record
Production Management	Planning, work orders, BOM, material consumption, production output	Production orders, consumption, finished/semi-finished stock
Sales Management	Customers, sales orders, invoicing, delivery and reporting	Sales order, invoice, delivery request/status
Gate Management	Vehicle/goods entry and exit, delivery verification	Gate entry/exit record linked to PO/GRN/SO/delivery
Finance integration	Payment request, authorization, supplier settlement	Approved payment, payment record, payment evidence

3. Enterprise-Level End-to-End Flow

Stage	Integrated process
1. Demand / Requirement	Department or Store identifies raw-material requirement or production demand.
2. Purchase Requisition	Storekeeper/requester creates PR with items, quantities, required date, store/location and justification.
3. PR Approval	Management/authorized approver reviews and approves, rejects or returns the PR.
4. Purchase Order	Purchasing converts the approved PR into a PO, selects supplier, confirms quantities/prices/terms and sends PO.
5. Supplier Delivery	Supplier delivers materials against the PO. Gate Management records vehicle and delivery entry.
6. Inspection / Laboratory	QC/Laboratory records tests and compares results with material specifications.

7. Accept / Reject	Accepted quantity proceeds to receiving. Rejected/nonconforming material is blocked and handled through return/replacement.
8. Goods Receipt	Storekeeper records accepted goods and generates GRN. Inventory is updated for the accepted quantity.
9. Payment Request	Finance creates a supplier Payment Request using approved PO, GRN and supplier invoice.
10. Payment Approval	Authorized approver reviews the payment request and approves/rejects/returns it.
11. Supplier Payment	Finance executes the approved payment by configured method such as bank transfer or cheque and stores payment evidence.
12. Close / Reconcile	Procurement, inventory, quality and finance records are linked and the transaction becomes auditable.

4. Detailed Cross-Module Flow

4.1 Procurement to Payment

1. Supplier Master → Supplier registration captures basic information, contacts, address, financial information, and product/service information.
2. PR → Requester selects required items and quantities and submits the requisition.
3. Approval → PR appears on requester/approver dashboards; authorized management takes the approval action.
4. PO → Purchasing creates the PO from the approved PR and sends it to the supplier.
5. Delivery → Supplier arrives with materials and delivery documentation; Gate Management logs the arrival and verifies the delivery reference.
6. Quality → Inspection/Laboratory records tests and results. Material status becomes Accepted or Rejected/Blocked.
7. Receiving → Storekeeper receives only accepted quantity and creates GRN. The source requirements identify Goods Receiving and subsequent stock balance maintenance.
8. Inventory → Accepted quantity updates stock balances for the relevant item and storage location.
9. Payment Request → Finance requests payment on behalf of the supplier using PO, GRN and supplier invoice.
10. Approval → Payment request is presented to the authorized approver for action.
11. Settlement → Finance records supplier payment and uploads supporting payment evidence/deposit slip where applicable.

4.2 Production Flow

12. Sales demand / forecast / internal production requirement creates or contributes to a production plan.
13. Production planning creates Work Orders based on the required product, quantity, date and BOM.
14. ERP checks raw-material availability in Inventory.
15. Materials are issued/reserved against the Work Order.
16. Production consumes materials and records actual consumption, waste/scrap and output.
17. Quality checks may be performed on production output according to configured controls.
18. Accepted finished/semi-finished goods are received into inventory.
19. Production performance, material variance and waste are reported.

4.3 Sales-to-Delivery Flow

20. Customer registration and master-data validation.
21. Sales Order creation with products, quantities, prices, delivery requirements and customer details.
22. Sales approval according to configured authority limits.
23. Inventory availability check and reservation/allocation where required.
24. Delivery preparation and Gate Management vehicle/dispatch control.
25. Goods issue/delivery confirmation updates inventory.

26. Invoice generation and sales reporting.
27. Delivery completion and document/audit closure.

5. Module-to-Module Integration Matrix

From	To	Integration event	Data passed
Procurement	Inventory	Approved PO / expected receipt	PO, supplier, item, ordered qty, delivery location
Procurement	Gate	Supplier delivery	PO, supplier, vehicle, delivery reference
Gate	Quality	Vehicle/material arrival	Gate entry, PO, supplier, delivery details
Quality/Lab	Inventory	Inspection accepted	Item, batch/lot, accepted qty, quality status
Quality/Lab	Procurement	Inspection rejected	Rejected qty, reason, test results, disposition
Inventory	Finance	GRN completed	GRN, accepted qty, PO reference, receipt status
Procurement/Inventory	Finance	Payment eligibility	PO + GRN + invoice references
Finance	Procurement	Payment completed	Payment reference, amount, date, evidence
Inventory	Production	Material availability	On-hand/reserved/blocked quantities
Production	Inventory	Consumption/output	Material consumption, waste, output quantities
Sales	Inventory	Order/delivery	SO, allocated qty, issued qty
Sales	Gate	Dispatch	Delivery reference, vehicle, customer/order
Gate	Inventory	Exit confirmation	Dispatch/GRN/SO reference, vehicle and exit timestamp

6. Core Master Data

Master	Important fields / controls
Supplier	Basic information, contacts, addresses, financial information, supplied products/services, active/inactive status
Item	Item code, description, category, unit, specifications, reorder levels, lot/batch requirements, quality requirements
Warehouse	Warehouse identity, location, responsible unit, status
Store	Store identity, warehouse relationship, responsible storekeeper, status
Silo	Silo identity, capacity, material compatibility, current balance/status
Customer	Customer profile, contacts, addresses, financial/credit information
BOM	Finished product, component items, quantities, units, version/effectivity
Quality Specification	Material/item, test parameters, acceptable ranges/criteria, sampling requirements
Vehicle	Plate/identifier, supplier/customer/driver relationship where applicable
Users/Roles	Requester, Storekeeper, Management Approver, Purchasing, QC/Lab, Finance Officer, Finance Approver, Gate Officer, Production, Sales, Administrator

7. Document and Transaction Chain

Document	Created by / source	Depends on	Next use
Supplier Registration	Procurement/Admin	Supplier master data	Supplier selection
Purchase Requisition (PR)	Requester/Storekeeper	Requirement	Approval and PO creation
Purchase Order (PO)	Purchasing	Approved PR + supplier	Supplier delivery and receiving
Gate Entry	Gate Officer	PO/delivery reference	Delivery verification
Inspection / Test Record	QC/Laboratory	PO/delivery/material	Accept/reject decision
Goods Receipt Note (GRN)	Storekeeper	Accepted material	Inventory and payment eligibility
Stock Ledger Entry	Inventory	GRN / issue / adjustment / production	Current balances and reporting
Supplier Invoice	Supplier / Finance	Supplier transaction	Payment request
Payment Request	Finance Officer	PO + GRN + invoice	Approval and settlement
Payment Record	Finance	Approved payment request	Supplier settlement/audit
Delivery / Dispatch	Sales/Store/Gate	Sales Order + inventory availability	Customer fulfillment
Work Order	Production	Plan + BOM	Material issue and production output

8. Decision and Exception Flows

Exception	System behavior
PR rejected/returned	Return to requester with reason; no PO may be created until approval.
Supplier unavailable/inactive	Purchasing selects another approved supplier or returns the PO workflow for correction.
Partial delivery	Record actual delivered quantity; leave outstanding quantity against PO for later delivery.
Inspection failed	Keep rejected quantity blocked; record reason/test result and initiate return, replacement or approved disposition.
Quantity mismatch	Do not silently increase stock. Require variance handling/authorization before final receipt.
Damaged goods	Record damaged quantity separately and prevent it from becoming unrestricted available stock.
Invoice mismatch	Hold payment request until PO, GRN and invoice discrepancies are resolved or formally approved.
Payment rejected	Return to Finance with reason; preserve the audit trail and prevent duplicate settlement.
Stock adjustment	Require reason, authorization and immutable transaction history.
Gate mismatch	Prevent or flag delivery/dispatch when required PO/SO/GRN references cannot be verified.

9. Key ERP Controls

- Every transaction must have a unique reference number and lifecycle status.
- Approval actions must record user, role, timestamp, decision and reason where applicable.
- A GRN must not exceed the allowable PO quantity without an explicit authorized variance process.
- Rejected/blocked material must not be available for normal production consumption.
- Payment eligibility should require traceable PO, GRN and supplier invoice references.
- Stock changes should be transaction-driven and auditable rather than direct balance edits.
- Stock adjustment must require reason and authorization.
- Role-based access control must restrict who can create, approve, receive, adjust, pay and administer records.
- Important master data changes should be audited.
- Documents and supporting evidence should remain linked to their originating transactions.

- Duplicate payment prevention should be enforced using supplier/invoice/payment references and transaction status.
- Gate entry and exit should be time-stamped and linked to the relevant business transaction.

10. Suggested Status Lifecycles

Transaction	Lifecycle
PR	Draft → Submitted → Under Review → Approved / Rejected / Returned → Converted to PO
PO	Draft → Submitted → Approved → Sent to Supplier → Partially Received / Fully Received → Closed / Cancelled
Delivery	Expected → Gate Entered → Inspection Pending → Accepted / Rejected / Blocked → Received / Returned → Gate Exited
Inspection	Pending → Sampled → Testing → Passed / Failed → Approved / Rejected
GRN	Draft → Pending Confirmation → Confirmed → Posted / Cancelled
Payment Request	Draft → Submitted → Under Review → Approved / Rejected / Returned → Paid / Cancelled
Work Order	Planned → Released → Material Issued → In Production → Completed / Cancelled
Sales Order	Draft → Submitted → Approved → Allocated → Dispatched → Delivered / Cancelled

11. ERP Reporting and Visibility

- Procurement: PR aging, PO status, supplier performance, outstanding orders, delivery performance.
- Inventory: current stock by warehouse/store/silo, stock movement, stock adjustment, aging, reorder alerts, blocked/rejected stock.
- Quality: inspection history, test results, rejected quantities, supplier quality trends, pending inspections.
- Production: work-order status, material consumption versus BOM, output, waste, downtime/performance indicators.
- Sales: order status, invoice status, delivery status, customer activity and sales reporting.
- Gate: vehicle entries/exits, pending deliveries, delivery verification exceptions.
- Finance: payment requests pending approval, approved/unpaid requests, paid suppliers, payment history and supporting evidence.
- Management: end-to-end procurement-to-payment visibility, inventory exposure, production status, sales fulfillment and exception dashboards.

12. Recommended ERP Architecture View

The ERP should be treated as one integrated transaction platform rather than six disconnected applications. Each module owns its business rules and records while sharing controlled master data and transaction references. The integration model should preserve a single chain of traceability from requirement → PR → PO → delivery → inspection → GRN → stock → invoice → payment.

Layer	Responsibility
Presentation	Role-specific dashboards, forms, approvals, operational work queues, reports
Application / Domain	Procurement, Inventory, Quality, Production, Sales, Gate and Finance business rules
Workflow	Approval routing, lifecycle/status transitions, notifications, exception handling
Integration	Internal domain events/APIs and external integrations such as payment/accounting where applicable

Data	Master data, transactional records, stock ledger, quality records, documents and audit history
Security	RBAC, authorization rules, segregation of duties, audit trail, authentication and session controls
Infrastructure	Application services, database, file/object storage, background jobs, monitoring, backups and recovery

13. End-to-End Traceability Example

A single raw-material delivery should be traceable using linked references: Supplier → PO → Gate Entry → Inspection/Test Record → Accepted Quantity → GRN → Stock Ledger → Supplier Invoice → Payment Request → Approval → Payment Record. The same material can then be traced forward from stock to Work Order → Material Consumption → Production Output → Finished Stock → Sales Order → Delivery → Customer.

14. Source Alignment and Design Notes

The uploaded reference explicitly documents supplier registration, PR, PO, raw-material inspection, raw-material store receiving, item/warehouse/store/silo masters, goods receiving, background stock-balance maintenance, stock adjustment, payment request, payment approval and supplier payment. The Scope of Work supplied in the request adds the broader Inventory, Production, Sales and Gate domains and their intended integrations.

Where the Scope of Work does not define detailed approval matrices, accounting rules, tax treatment, production planning rules, dispatch rules, laboratory standards, or payment limits, this document presents them as proposed ERP design controls rather than as confirmed business requirements.

Reference: uploaded 18-page requirements PDF. The procurement-to-payment process is stated on page 1; supplier registration is covered on pages 3–5; PR and PO on pages 6–7; inspection on page 8; item/warehouse/store/silo and goods receiving on pages 9–13; stock balance and adjustment on pages 14–15; and payment request, approval and supplier payment on pages 16–18.

15. Business Questions to Finalize Before Implementation

28. What exact floor-manufacturing products and production stages will be managed?
29. Are raw materials batch/lot controlled, and which materials require mandatory laboratory testing?
30. Which quality standards and acceptable test ranges apply to each raw-material type?
31. Can partial deliveries and partial GRNs be posted against one PO?
32. What quantity/price variance tolerances are allowed between PR, PO, GRN and invoice?
33. What is the approval matrix by amount, department, item category and transaction type?
34. Should Finance use three-way matching as a hard control before payment?
35. How should rejected material be returned, replaced or disposed?
36. How many warehouses, stores and silos exist, and can stock move between them?
37. Which production transactions require material reservation, issue, return and scrap?
38. What are the sales-to-delivery approval and credit-control rules?
39. What gate documents must be checked for inbound and outbound movements?
40. Which accounting/ERP/payment systems must integrate with this platform?
41. What are the required tax, invoice, withholding and payment rules?
42. What audit, retention, backup and disaster-recovery requirements apply?

16. Overall ERP Process Summary

Demand initiates procurement or production. Procurement establishes the supplier commitment through PR and PO. Gate Management records physical movement. Quality/Laboratory validates incoming raw materials. Inventory records only accepted stock through GRN and maintains the stock ledger. Finance uses the linked PO, GRN and invoice to control supplier payment. Production consumes controlled inventory and creates output. Sales converts customer demand into orders, invoices and deliveries. Gate Management verifies physical dispatch. Across all modules, workflow, authorization, auditability and linked transaction references provide the common ERP control layer.